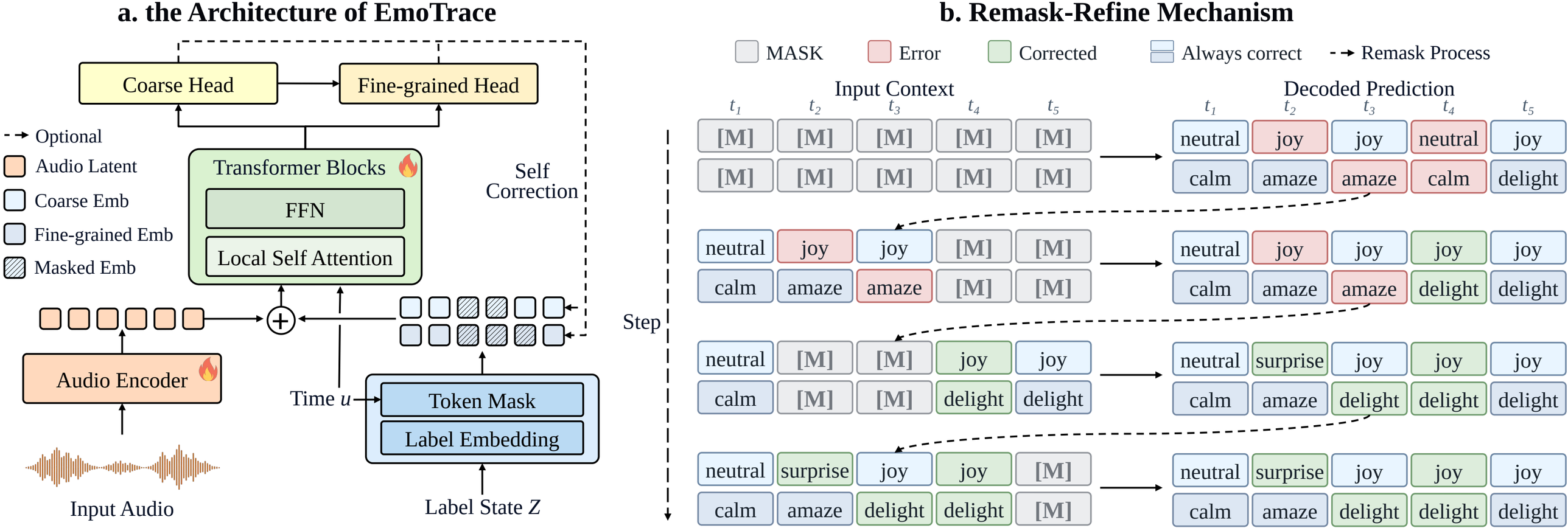

a. the Architecture of EmoTrace
Coarse Head
Fine-grained Head
Optional
Audio Latent
Coarse Emb
Fine-grained Emb
Masked Emb
Transformer Blocks
FFN
Local Self Attention
Self Correction
Audio Encoder
Time u
Token Mask
Label Embedding
Input Audio
Label State Z
b. Remask-Refine Mechanism
MASK
Error
Corrected
Always correct
Remask Process
Input Context
Decoded Prediction
Step
t1 t2 t3 t4 t5
[M] [M] [M] [M] [M]
[M] [M] [M] [M] [M]
neutral joy amaze amaze calm delight
neutral joy joy [M] [M]
calm amaze amaze [M] [M]
neutral joy joy joy joy
calm amaze amaze delight delight
neutral [M] [M] joy joy
calm [M] [M] delight delight
neutral surprise joy joy joy
calm amaze delight delight delight
neutral surprise joy joy [M]
calm amaze delight delight [M]
neutral surprise joy joy joy
calm amaze delight delight delight